

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=cottage_cheese

retailer=silpo_novus

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	nan	nan	nan	nan	nan	nan	ambiguous	0.0	cottage_cheese	silpo_novus	promo_controlled	nan	nan
prozorro	1.1483227108295005e-10	0.01	4.589982408975472e-17	0.1	4.813752592550519e-10	0.1	I(1)	0.0	cottage_cheese	silpo_novus	promo_controlled	nan	nan
retail	7.053635234015645e-08	0.055181023671470035	0.0008960877939895696	0.1	4.7830375146070525e-06	0.1	I(0)	0.0	cottage_cheese	silpo_novus	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	cottage_cheese	silpo_novus	promo_controlled	nan	0.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	cottage_cheese	silpo_novus	promo_controlled	0.009936705828205612	42.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	cottage_cheese	silpo_novus	promo_controlled	nan	0.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
cottage_cheese	silpo_novus	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	42	-0.40662631250165315	-1.685428485355544	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=0.02475; b...	Primary chain is strictly...
cottage_cheese	silpo_novus	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	40	-0.04779905990700531	-0.5494671968185121	-0.9346201147084174	1.5479768355429064e-10	0.8868176751444053	0.32025906973649765	nan	ok	0	Wald short/long asymmetry...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
ARDL	prozorro_to_retail	retail	prozorro	0.10074476960434399	0.4559102488077962	0.7986167356499632	0
NARDL	prozorro_to_retail	retail	prozorro	0.13802457343734706	0.7478438770300309	2.6950380978724736e-06	0

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2025-04-14	nan	138.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-16	nan	118.8	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-18	nan	108.9	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-21	nan	120.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-22	nan	99.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-23	nan	107.08	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-24	nan	160.2	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-29	nan	156.06	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-30	nan	226.25	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-01	nan	82.5	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-02	nan	118.08000000000001	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-05	nan	101.29499999999999	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-06	nan	115.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-07	nan	258.2	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-08	nan	145.4	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-09	nan	101.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-11	nan	89.4	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-12	nan	168.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-13	nan	131.34	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-14	nan	75.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-15	nan	152.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-16	nan	145.2	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-19	nan	98.75	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-20	nan	159.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-21	nan	134.04	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-22	nan	139.98000000000002	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-23	nan	107.55	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-26	nan	168.6	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-27	nan	156.06	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-29	nan	150.5	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-30	nan	165.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-02	nan	135.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-03	nan	122.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-05	nan	142.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-06	nan	90.52000000000001	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-09	nan	60.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-10	nan	112.98	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-11	nan	93.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-12	nan	108.0	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-13	nan	71.5	nan	cottage_cheese	silpo_novus	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
prozorro_to_retail	1	-0.08078408601258956	
prozorro_to_retail	2	-0.06454950488841439	
prozorro_to_retail	3	-0.1933334718038668	
prozorro_to_retail	4	0.18827209925822697	
prozorro_to_retail	5	0.14536351827047989	
prozorro_to_retail	6	-0.2545432657888236	
prozorro_to_retail	7	-0.12508455517529882	
prozorro_to_retail	8	0.08631606508396195	
prozorro_to_retail	9	0.1355966845634848	
prozorro_to_retail	10	0.2139347251618982	
prozorro_to_retail	11	-0.009468860585183389	
prozorro_to_retail	12	0.2419733955614274	
prozorro_to_retail	13	0.043158430923380946	
prozorro_to_retail	14	-0.1931909836115445	
prozorro_to_retail	15	-0.28301809398285815	
prozorro_to_retail	16	-0.07939066612951604	
prozorro_to_retail	17	-0.08985615650558762	
prozorro_to_retail	18	0.018032174074130422	
prozorro_to_retail	19	0.23940024361697732	
prozorro_to_retail	20	0.23013991101982256	
prozorro_to_retail	21	-0.1517583037018874	
prozorro_to_retail	22	-0.19380051895183734	
prozorro_to_retail	23	-0.09515676503663541	
prozorro_to_retail	24	0.07460101451181338	
prozorro_to_retail	25	0.20413401238475085	
prozorro_to_retail	26	-0.1023782790163822	
prozorro_to_retail	27	-0.12197770872206383	
prozorro_to_retail	28	0.15270196529852897	
prozorro_to_retail	29	0.3172406495233786	
prozorro_to_retail	30	0.19856144431286887	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
cottage_cheese	silpo_novus	promo_controlled	daily	ambiguous	I(1)	I(0)	0	190	55	0	42	0	Low pair overlap in main ...

Table: NARDL_Multipliers

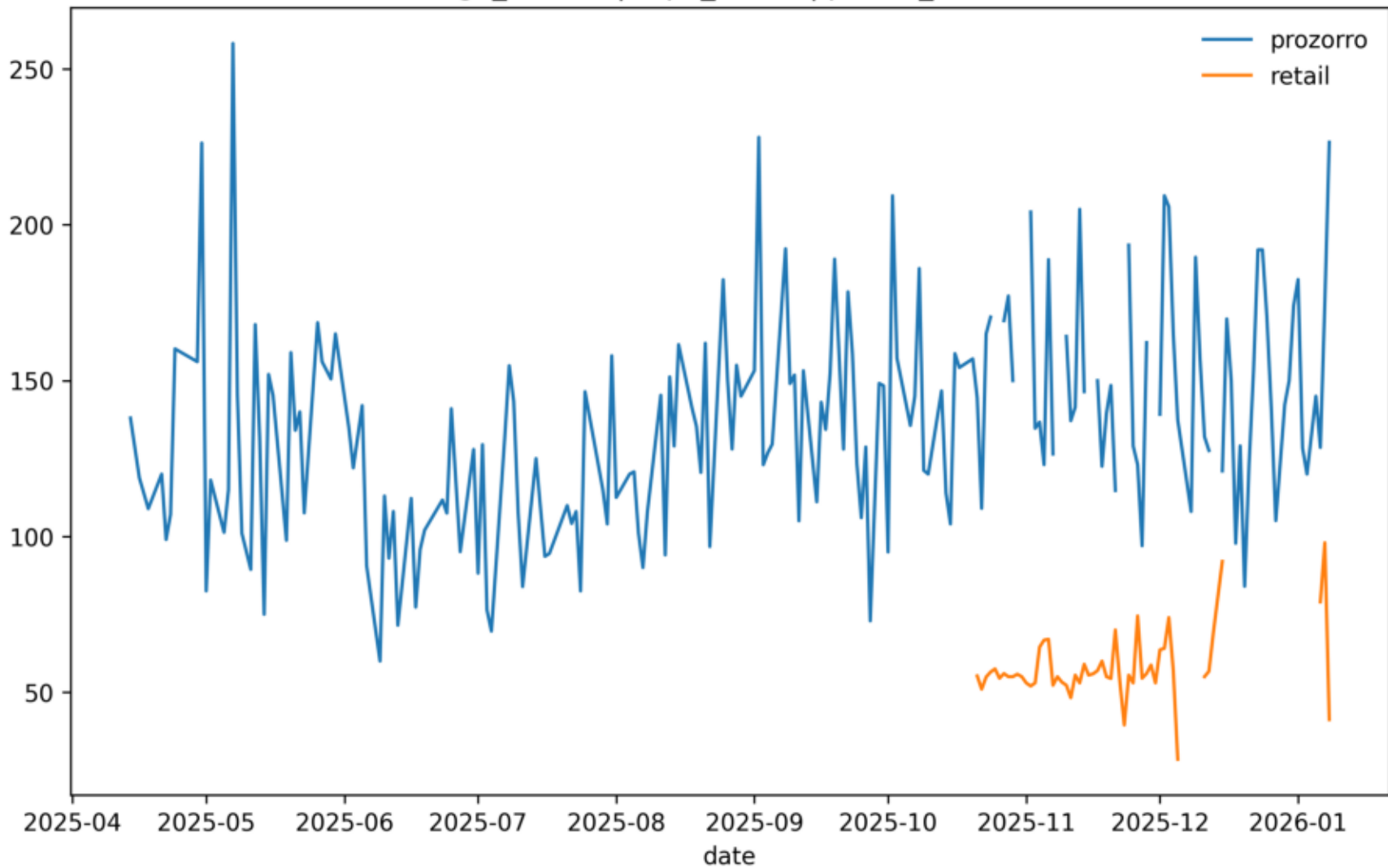
horizon	mult_pos	mult_neg	link
0	-0.25026320175044375	-0.20246414184343844	prozorro_to_retail
1	-0.23767528150629358	-0.19228045342260974	prozorro_to_retail
2	-0.2383084378591444	-0.192792679977437	prozorro_to_retail
3	-0.23827659090115771	-0.1927669156342415	prozorro_to_retail
4	-0.23827819276257542	-0.1927682115478837	prozorro_to_retail
5	-0.23827811219098669	-0.19276814636507833	prozorro_to_retail
6	-0.23827811624363496	-0.1927681496436904	prozorro_to_retail
7	-0.2382781160397919	-0.19276814947878038	prozorro_to_retail
8	-0.23827811605004495	-0.19276814948707516	prozorro_to_retail
9	-0.23827811604952925	-0.19276814948665794	prozorro_to_retail
10	-0.23827811604955518	-0.19276814948667892	prozorro_to_retail
11	-0.23827811604955387	-0.19276814948667786	prozorro_to_retail
12	-0.23827811604955396	-0.19276814948667792	prozorro_to_retail
13	-0.23827811604955396	-0.19276814948667792	prozorro_to_retail
14	-0.23827811604955396	-0.19276814948667792	prozorro_to_retail
15	-0.23827811604955396	-0.19276814948667792	prozorro_to_retail
16	-0.23827811604955396	-0.19276814948667792	prozorro_to_retail
17	-0.23827811604955396	-0.19276814948667792	prozorro_to_retail
18	-0.23827811604955396	-0.19276814948667792	prozorro_to_retail
19	-0.23827811604955396	-0.19276814948667792	prozorro_to_retail
20	-0.23827811604955396	-0.19276814948667792	prozorro_to_retail

Table: VECM_IRF

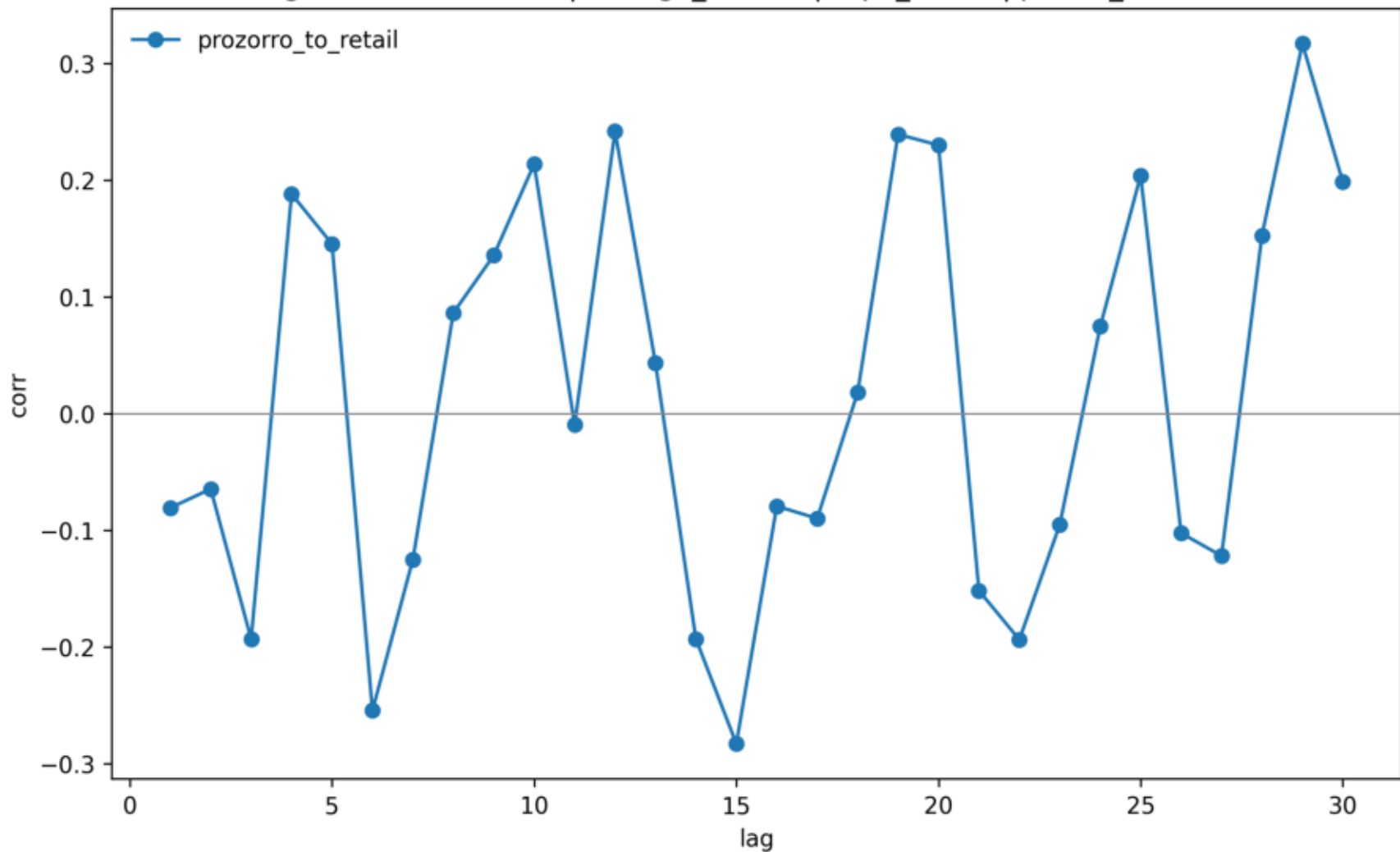
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time_series_promo_controlled.png

cottage_cheese | silpo_novus | promo_controlled



Lag-Correlation Profile | cottage_cheese | silpo_novus | promo_controlled



NARDL Dynamic Multipliers | cottage_cheese | silpo_novus | promo_controlled

